

Experiment 4: Iterative Statements (Loops)

Objective: To solve problems that require repetitive tasks using for, while, and do-while loops.

Questions:

- 1. Write a C program to count the number of digits in a number using while loop.**

Sample Input:

Enter a number: 12345

Sample Output:

Number of digits: 5

- 2. Write a C program to print all Armstrong numbers between 1 to n.**

Sample Input:

Enter the value of n: 500

Sample Output:

Armstrong numbers between 1 and 500 are: 1, 153, 370, 371, 407

- 3. Write a C program to input a number and check whether the number is a Perfect number or not.**

Sample Input:

Enter a number: 6

Sample Output:

6 is a Perfect number.

- 4. Write a program to print the Fibonacci series up to n terms using do-while loop.**

Sample Input:

Enter the number of terms: 10

Sample Output:

Fibonacci Series: 0 1 1 2 3 5 8 13 21 34

- 5. Write a program to find the GCD (Greatest Common Divisor) of two numbers.**

Sample Input:

Enter two numbers: 60 48

Sample Output:

GCD of 60 and 48 is 12.

- 6. Write a C program to check whether a given number is a palindrome using for loop.**

Sample Input:

Enter a number: 1221

Sample Output:

1221 is a Palindrome Number

7. Write a C program to print a hollow square star pattern using a for loop.

Sample Input:

Enter the number of rows: 5

Sample Output:

```
*****
*   *
*   *
*   *
*****
```

8. Write a C program to print reverse Pyramid or reverse equilateral triangle star pattern series using for loop.

Sample Input:

Enter the number of rows: 5

Sample Output:

```
* * * * *
 * * * *
  * * *
   * *
    *
   
```

9. Write a C program to print Floyd's triangle.

Sample Input:

Enter the number of rows: 4

Sample Output:

```
1
2 3
4 5 6
7 8 9 10
```

10. Write a C program to print Pascal's triangle using loops.

Sample Input:

Enter the number of rows: 5

Sample Output:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

(Each number is generated from the previous one using: $next = current \times \frac{(row - column)}{(column + 1)}$)